Section 1.7 — Absolute Value Functions

Math 103: College Algebra

Evaluating Absolute Values

Evaluate each expression.

1. $|2-3| + \dfrac{|5|}{|-4|}$
2. $-|-5| \cdot |7-1|$
3. $2 \cdot -|6+3| - |-2 \cdot 6|$
4. $-\frac{1}{3}|2-8| + 4$

Identifying Absolute Value Functions

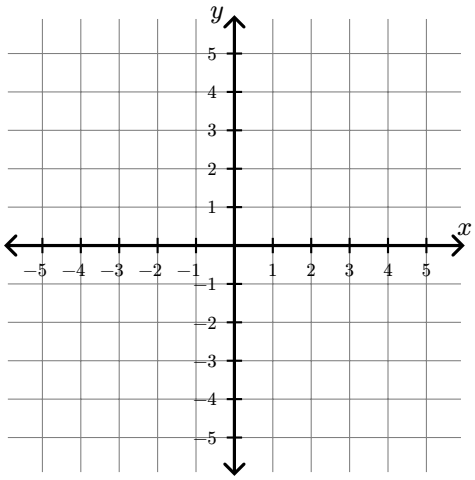
Determine whether each of the following is an absolute value function.

5. $f(x) = |x-1| + 3$
6. $g(x) = -(2x+4) + 6$
7. $h(x) = \left|x^2 - 3x + 2\right|$
8. $k(x) = \frac{1}{2}|4-x| + 1$
9. $r(x) = \dfrac{3}{|4x-1|}$
10. $s(x) = \sqrt{|2x-1|}$
11. $t(x) = -|2x-3| - \frac{3}{4}$
12. $u(x) = \left|\frac{1}{3}x - 5\right| - \dfrac{4}{x^2}$

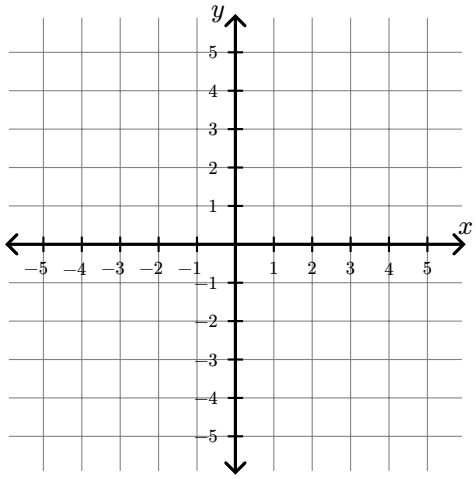
Graphing Absolute Value Functions

Graph each absolute value function. Then, looking at the graph, state the domain and range of each function.

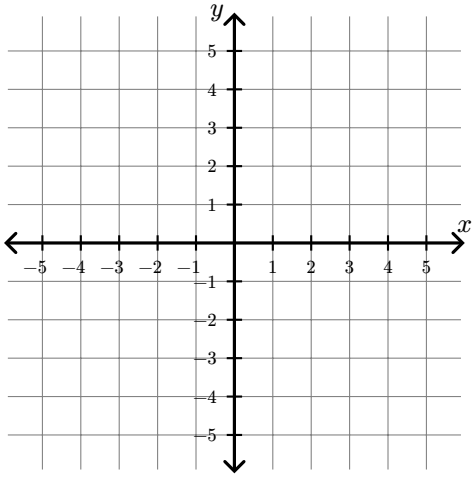
13. $f(x) = \frac{2}{3}|x|$



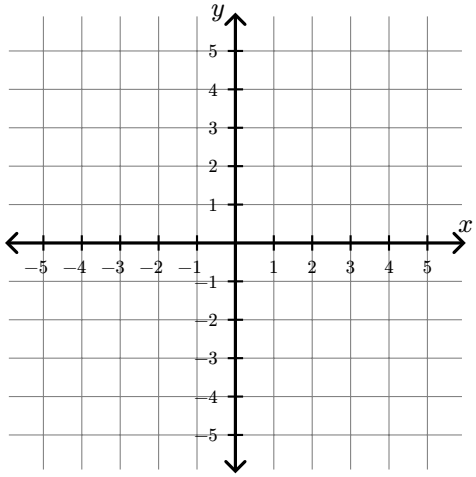
14. $g(x) = 2|x-4| - 2$



15. $h(x) = -\frac{3}{4}|x| + 1$



16. $k(x) = -\frac{3}{2}|x+1| - 1$



Range and Domain of an absolute value function

Find the domain and range of each function.

17. $f(x) = |x - 2|$

18. $g(x) = 2\,|x| + 4$

19. $h(x) = -3\,|x - 1| + 3$

20. $k(x) = \frac{1}{3}\,|2x + 1| - 4$